

Diff-DDoS: Realistic Cyber-Physical Attack Synthesis and Robust Detection for 5G-Enabled CPS Using Tabular Diffusion Models

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Abstract—Deep learning-based DDoS detectors for 5G-enabled cyber-physical systems face two challenges: scarce real-world labeled attack data and the unreality of naive synthetic substitutes, which limit robustness against adaptive adversaries. Detectors trained on hand-crafted attacks with fixed scaling multipliers degrade catastrophically—F1-score drops of $\sim 47\%$ to 100% depending on scenario—when confronted with realistic, distribution-preserving samples. We propose *Diff-DDoS*, a three-phase framework for realistic attack synthesis and robust detection using tabular diffusion models. Phase 1 establishes a baseline CNN cell-level detector on spatiotemporal grids derived from call detail records (CDRs). Phase 2 trains a tabular denoising diffusion probabilistic model (TabDDPM) on normal CDR aggregates to generate realistic attacks, exposing detector vulnerabilities. Phase 3 introduces adversarial diffusion training (ADT), which applies inverse classifier guidance to iteratively generate hard yet distribution-preserving adversarial samples until the detector converges. On a real-world Milano CDR dataset across SMS-flooding, silent-call, Internet-signaling, and blended scenarios, ResNet50 with ADT recovers F1-scores

of 79.62% (silent-call), 100% (Internet), and 92.79% (blended) while retaining near-perfect multiplier-test F1 on most scenarios. We further benchmark ADT against gradient-based adversarial training, conditional tabular GAN (CTGAN), and fixed-multiplier baselines under identical iterative training and validation-based threshold calibration; after calibration, ADT reaches 100% SMS F1 versus 47.3% for CTGAN and attains silent-call F1 on par with the strongest gradient-based adversarial-training baseline. SMS flooding remains challenging for lightweight architectures, though calibration enables near-perfect ResNet50 detection. These results establish tabular diffusion models as a practical tool for stress-testing and hardening intrusion detectors in data-scarce 5G cyber-physical deployments.

Index Terms—5G networks, adversarial training, cyber-physical systems, DDoS attacks, diffusion models.

I. INTRODUCTION

FIFTH-generation (5G) mobile networks are becoming a core communication substrate for cyber-physical systems (CPSs), including industrial monitoring, transportation, and energy services. This convergence enlarges the attack surface and creates new pathways for distributed denial-of-service (DDoS) attacks that threaten both network availability and physical processes [1]–[4], requiring multi-layer AI-enabled protection [5], [6]. Recent threat-landscape reports rank DDoS attacks among the most prominent cybersecurity threats in mobile and Internet of Things (IoT)-connected environments [7].

Detecting *stealthy* cellular attacks is particularly difficult. Silent-call abuse, SMS flooding, and Internet-signaling (signaling storm) attacks stress control-plane procedures rather than only consuming user-plane bandwidth [8]–[11], and signaling overload and layer-3 abuse remain credible threats in modern cellular architectures [10], [11].

At the monitoring layer, call detail record (CDR) aggregates capture cell-level temporal deviations in SMS, call, and Internet activity. Hussain *et al.* [4] showed they can be organized as $9 \times 9 \times 3$ spatiotemporal grids and classified by convolutional neural networks (CNNs) with strong performance on multiplier-based synthetic attacks. However, those multiplier attacks use fixed, large feature multipliers and are therefore easier to separate from normal traffic than realistic adversarial behavior.

This mismatch exposes a broader weakness: models trained on stylized attacks degrade sharply under distribution

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shift [12], [13]. In cellular settings the problem is amplified by scarce, privacy-sensitive attack labels; naive perturbations can break traffic semantics [2], [4], [12], [13], leaving detectors brittle against attacks that stay near the normal traffic manifold.

To address this gap, we propose Diff-DDoS, a three-phase framework using tabular diffusion models to synthesize realistic attack samples and harden detectors against adaptive evasion. In **Phase 1**, we re-implement the CNN-based detector of [4] as a multiplier-based baseline (full pseudo-code in Supplementary Sec. S-A). In **Phase 2**, we train TabDDPM only on normal CDR aggregates to generate manifold-preserving attack samples, revealing the baseline detector's poor generalization. In **Phase 3**, we introduce *adversarial diffusion training* (ADT), which uses guided diffusion sampling to generate hard on-manifold attacks and iteratively retrains the detector.

The main contributions of this work are as follows:

- 1) We re-implement the cell-level CNN baseline of [4] and establish reference performance on multiplier-based attacks.
- 2) We demonstrate that detectors trained only on multiplier-based attacks can fail severely when evaluated on diffusion-generated attacks drawn from the normal CDR manifold.
- 3) We introduce ADT, which combines TabDDPM-based generation with inverse classifier guidance to create hard, realistic attack samples for iterative robust training.
- 4) We evaluate Diff-DDoS across SMS, silent-call, Internet, and blended scenarios using SimpleCNN and 50-layer residual network (ResNet50), highlighting ADT robustness gains, validation-based threshold calibration, and a per-scenario learnability analysis showing that SMS flooding is capacity-limited on SimpleCNN but recoverable on ResNet50.

Sec. II covers related work; Sec. III background and problem setup; Sec. IV methodology; Sec. V experimental setup and evaluation protocols; Sec. VI results; Sec. VII limitations and conclusion. Extended derivations and full result grids are in the Supplementary Material.

II. RELATED WORK

DDoS detection in 5G-enabled CPSs has advanced from rule-based to machine learning/deep learning (ML/DL) methods because signaling abuse, SMS flooding, and silent-call attacks can evade threshold defenses [2], [14]. Hussain *et al.* [4] showed CDR grids enable strong CNN performance on multiplier-based attacks; later work explored recurrent, federated, and quantum-inspired models [15]–[17].

CDR analytics also demonstrate useful spatiotemporal structure for traffic prediction and network monitoring [18]–[21]. Yet both DDoS detection and CDR analytics literatures rarely evaluate robustness to adaptive evasion close to normal traffic statistics [4], [15], [22], [23].

Deep intrusion detection systems (IDSs) degrade under distribution shift and adversarial manipulation [24]–[26]. Conditional tabular generative adversarial network (CTGAN) improves data augmentation but suffers from instability [27];

TabDDPM offers richer tabular distributions and higher-fidelity synthesis [28], [29]. Gradient-based adversarial perturbations generate off-manifold samples [12], [13], [30], [31], while diffusion-based approaches show growing promise [32]–[34].

III. BACKGROUND AND PROBLEM SETUP

A. CDR Grid Representation and Detection Task

Each CDR snapshot is represented as a grid image $x \in \mathbb{R}^{9 \times 9 \times 3}$, where the three channels correspond to aggregated SMS-out, Call-out, and Internet traffic for each of the $9 \times 9 = 81$ cells in a selected Milano [35] subgrid over one 10-minute interval within the 11:00–14:00 window. The detection task is formulated as a multilabel cell-level classification problem: for each grid image x , a CNN-based detector $\mathcal{M}$ with learnable weights θ outputs a vector

$$\mathcal{M}(x; \theta) = \hat{Y} \in (0, 1)^{81},$$

where each component $\hat{y}_k \in (0, 1)$ is the sigmoid-activated predicted attack probability for cell $k \in \{1, \dots, 81\}$. The ground-truth label $Y \in \{0, 1\}^{81}$ assigns $Y_k = 1$ if cell k is under attack and $Y_k = 0$ otherwise. During adversarial diffusion training (Phase 3), θ is updated iteratively across rounds, so the detector after ADT round r for attack scenario s is denoted $\mathcal{M}(\cdot; \theta_s^r)$ (during round- r sampling, guidance uses θ_s^{r-1} ; Sec. IV-D). We write $f_\theta \equiv \mathcal{M}(\cdot; \theta)$ in algorithms when shorthand is needed.

The CDR-based attack synthesis of [4] applies scenario-specific multipliers to $|\mathcal{C}| = 40$ randomly selected cells, labeling them $Y_k = 1$ (see Supplementary Algorithm S2). Four scenarios are evaluated throughout: *SMS flooding* (elevated SMS-out), *silent-call* (suppressed outgoing calls), *Internet-signaling* (elevated Internet, approximating a signaling storm [4]), and *blended* (simultaneous SMS/Internet elevation and call suppression). Phase 1 uses fixed multipliers ($169.5 \times$ SMS, $-33.1 \times$ call clipped at zero, $43.3 \times$ Internet); these stylized attacks motivate the diffusion-based robustness evaluation and are *not* used to generate attacks in Phases 2–3 (Phase 2 test attacks are TabDDPM-generated; see Sec. IV-C). The preprocessing pipeline (62 CDR files, 11:00–14:00 aggregation, Num2Dims($\cdot$) row-major cell indexing) is given in Supplementary Algorithms S1–S2.

B. Denoising Diffusion and Classifier Guidance

We use standard denoising diffusion probabilistic model (DDPM) notation [28], [36]: a fixed forward schedule corrupts clean data x_0 into noisy samples x_t over T steps; a learned reverse denoiser ϵ_ψ (weights ψ , distinct from detector θ) predicts the injected noise. Our TabDDPM instantiation on 3D CDR aggregates is summarized in Sec. IV-C.1; full equations are in Supplementary Sec. S-B. DDPMs have been extended from images [37] to tabular data [28]; Diff-DDoS builds on the latter.

TABLE I: Main-text notation summary.

Symbol	Meaning
$\mathcal{M}(\cdot; \theta), f_\theta$	Multilabel CNN detector (81 sigmoid outputs per grid)
$\Xi_s^r, \Xi_s^{r,\text{orig}}, \Xi_s^{r,\text{Tab}}$	Metric bundles (Acc, Prec, Rec, F1, FPR) for attack scenario s at ADT round r ; <i>not</i> the detector
$\theta, \theta_s^*, \theta_s^r$	Detector learnable weights
$\epsilon_\psi, \phi_{\text{TabDDPM}}$	TabDDPM denoiser network (weights ψ ; distinct from θ)
$\phi_s : \mathbb{R}^3 \rightarrow \mathbb{R}^3$	Mild post-sample scenario scaling on 3D CDR aggregates (Phase 2)
$\text{MapToGrid}(\tilde{\mathbf{x}}, B)$	Copy background B , write aggregate $\tilde{\mathbf{x}}$ into attacked cells $\mathcal{C}$, set $Y_k = 1$ on $\mathcal{C}$
p_{attack}	Mean detector score over attacked cells $\mathcal{C}$ during inverse guidance

Classifier guidance [37] steers the reverse process via classifier gradients:

$$\nabla_{x_t} \log p(x_t | y) = \underbrace{\nabla_{x_t} \log p(x_t)}_{\text{unconditional score}} + s_{\text{cg}} \cdot \underbrace{\nabla_{x_t} \log p(y | x_t)}_{\text{classifier gradient}}, \quad (1)$$

where ∇_{x_t} is the gradient with respect to the noisy sample x_t , s_{cg} controls conditioning strength, and the classifier gradient biases samples toward target class y . Diff-DDoS extends this idea via *inverse* guidance in Phase 3 (Sec. IV-D), steering samples away from attack-like detector scores while preserving the normal CDR manifold [33], [34].

C. Notation Summary

Table I lists overloaded symbols used in the main text; the supplement provides an extended disambiguation table.

IV. METHODOLOGY

Fig. 1 summarizes the Diff-DDoS pipeline: **Phase 1** establishes a multiplier-trained CNN baseline; **Phase 2** trains TabDDPM on normal 3D CDR aggregates and exposes baseline vulnerability to manifold-preserving attacks; **Phase 3** applies adversarial diffusion training (ADT) with inverse classifier guidance to iteratively harden the detector. Offline attack synthesis is separated from online grid-level inference (Sec. VI-F).

A. Data Representations and Attack-Steering Mechanisms

Normal tabular training rows are $\mathcal{R}_{\text{norm}} = \{(\text{SMS_sum}, \text{CALL_sum}, \text{INTERNET_sum})\}$ extracted from raw CDR aggregates over the 9×9 subgrid; they are standardized with `StandardScaler` fit on $\mathcal{R}_{\text{norm}}$ only. `Num2Dims( $k$ )` maps cell index k to grid coordinates (h, w) in row-major order. `MapToGrid( $\tilde{\mathbf{x}}, B$ )` copies background grid B , writes aggregate $\tilde{\mathbf{x}}$ into attacked cells $\mathcal{C}$ via `Num2Dims( $\cdot$ )`, and sets $Y_k = 1$ on $\mathcal{C}$. Phase 2 *test* grids use $B \in \mathcal{A}$ (abnormal-template pool); Phase 3 ADT *training* grids use a zero background ($B = \mathbf{0}$): only cells in $\mathcal{C}$ receive $\tilde{\mathbf{x}}$; all others remain zero (Algorithm 2). Table II contrasts the three attack-steering mechanisms used across phases.

TABLE II: Attack-steering mechanisms across Diff-DDoS phases.

Mechanism	Phase	Action	Purpose
Fixed multipliers	1	$169.5 \times$ SMS, $-33.1 \times$ Call, $43.3 \times$ Internet	Stylized baseline
Heuristic steering	2	Mild γ ranges on target channel(s)	On-manifold vulnerability test
ϕ_s Inverse classifier guidance	3	Minimize p_{attack} toward p_{target}	Hard robust-training samples

B. Phase 1: Baseline Detector Training

The detector is a *multilabel* model with 81 sigmoid outputs per grid (Sec. III-A); Phase 1 trains on the union of normal grids (all-negative labels) and multiplier-attack grids (40 perturbed cells positive), without any TabDDPM-generated attacks. Phase 1 follows [4]; the full training procedure is in Supplementary Algorithm S3.

Data Preprocessing: Phase 1 uses the two-stage pipeline of Supplementary Algorithms S1–S2 (grid extraction and fixed-multiplier attack injection on $|\mathcal{C}| = 40$ cells; Table II). Attack labels are defined solely by synthesis; no statistical thresholding is applied.

Model Training: Two detector architectures are employed:

- **SimpleCNN:** A lightweight CNN with three convolution–pooling blocks followed by fully-connected layers.
- **ResNet50:** A deeper (50-layer) residual network adapted to $9 \times 9 \times 3$ inputs with an 81-way sigmoid output layer.

Both models are trained with a multilabel binary cross-entropy (BCE) loss averaged over N training images and all 81 cells:

$$\mathcal{L}_{\text{BCE}}(\theta) = -\frac{1}{N} \sum_{i=1}^N \frac{1}{81} \sum_{k=1}^{81} \left[y_{i,k} \log \hat{y}_{i,k} + (1 - y_{i,k}) \log(1 - \hat{y}_{i,k}) \right], \quad (2)$$

where $y_{i,k} \in \{0, 1\}$ is the ground-truth label and $\hat{y}_{i,k} = \mathcal{M}(x_i; \theta)_k \in (0, 1)$ is the sigmoid-activated predicted attack probability for cell k in image i , produced by detector $\mathcal{M}$ with learnable weights θ .

Predictions are binarized at $\tau = 0.5$; true positives, false positives, and false negatives (TP/FP/FN) are micro-averaged over $N_{\text{test}} \times 81$ cell decisions to compute Accuracy, Precision, Recall, F1 score, False Positive Rate (FPR), and receiver operating characteristic area under the curve (ROC–AUC) (Sec. VI-G; dataset splits in Sec. V).

C. Phase 2: TabDDPM-Based Vulnerability Demonstration

Phase 2 demonstrates that multiplier-trained detectors are vulnerable to realistic diffusion-based attacks. Let $\mathcal{X}_n$ denote the 781 normal-only grid images (614 train + 167 test) and $\mathcal{A}$ the 335 unperturbed abnormal-template grids produced by Supplementary Algorithms S1–S2 (fixed random seed 1; Sec. V). Let $\mathcal{X}_a^s$ denote the scenario-specific abnormal grid working pool ($N_{\text{abn}} = 335$) used in Phase 2 indexing.

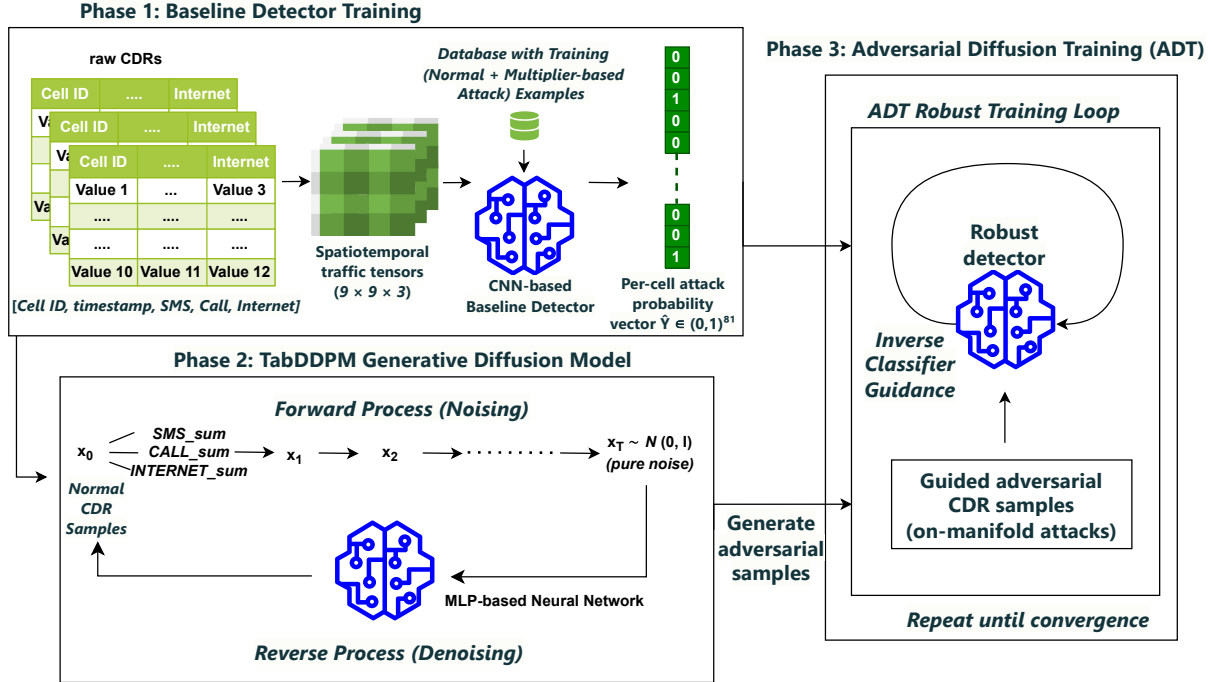


Fig. 1: Overview of the proposed Diff-DDoS framework. Phase 1 trains baseline detectors on multiplier-based attacks, Phase 2 uses TabDDPM to generate realistic attacks from normal CDR aggregates, and Phase 3 applies ADT to generate hard on-manifold samples for iterative retraining.

Training attacks in $\mathcal{X}_a^s$ (indices 0–166) remain multiplier-based, while test attacks (indices 167–334, $N_{\text{attack}}^{\text{test}} = 168$) are replaced with TabDDPM-generated grids. Let $\mathcal{X}_a^{s, \text{Tab}}$ denote the TabDDPM-based test set: the 335-image split combining 167 held-out normal test grids with 168 TabDDPM test-attack grids $\mathcal{X}_a^s[167:335]$ (i.e., $\mathcal{D}_{\text{test}}^{(s)}$ in Algorithm 1). Phase-2 F1 is $F1_{\text{Ph.2}}^{(s)}$ on this split with $\tau = 0.5$ (protocol P1; Table III); degradation is $\Delta_{F1}^{(s)} = (F1_{\text{Ph.1}}^{(s)} - F1_{\text{Ph.2}}^{(s)}) / F1_{\text{Ph.1}}^{(s)} \times 100\%$.

1) Tabular Diffusion Model for Aggregated CDRs: Attack synthesis operates on 3D CDR aggregates (SMS, CALL, INTERNET) per attacked-cell group, where scenario constraints are naturally expressed and realism is quantified (Sec. VI-C). A grid-image DDPM would couple unrelated cells and complicate per-channel control; we therefore model the low-dimensional aggregate manifold with TabDDPM [28] and map sampled aggregates back to $9 \times 9 \times 3$ grids via $\text{MapToGrid}(\cdot, B)$ (Sec. IV-A).

We adapt the DDPM framework [36] to tabular CDR features. Raw CDR rows are aggregated into a 3D feature vector $x_0 = (\text{SMS_sum}, \text{CALL_sum}, \text{INTERNET_sum}) \in \mathbb{R}^3$ per (cell, timeslot, day), which defines the normal CDR manifold. A fixed linear noise schedule ($T = 1000$, $\beta_{\min} = 10^{-4}$, $\beta_{\max} = 0.02$) governs the forward diffusion, and the denoiser ϵ_ψ —a multilayer perceptron (MLP) with hidden dimensions [256, 512, 256] that takes the concatenation of x_t and the scalar diffusion step t (broadcast to the feature dimension)—is trained with the standard ℓ_2 noise-prediction objective $\mathcal{L}_{\text{DDPM}}(\psi)$. The full forward/reverse equations, denoiser architecture, training objective, and posterior sampling are given in Supplementary Sec. S-B.

2) Realistic Attack Scenario Generation: TabDDPM is trained exclusively on normal aggregate CDR rows. Here $\mathcal{U}(a, b)$ denotes the continuous uniform distribution on $[a, b]$, and $\gamma_1, \gamma_2, \gamma_3$ are independent per-sample scaling factors. To generate realistic attacks for a given scenario s , we:

- 1) Run reverse diffusion (Supplementary Sec. S-B) to obtain an aggregate $\mathbf{x}^{(i)} = (\text{SMS}, \text{CALL}, \text{INTERNET}) \in \mathbb{R}^3$, applying lightweight heuristic steering on the target channel(s) every $\max(1, \lfloor T/10 \rfloor)$ timesteps (Table II; distinct from Phase 3 inverse classifier guidance and from post-sample ϕ_s in item 2).
- 2) Apply mild scenario-specific constraints $\tilde{\mathbf{x}}^{(i)} \leftarrow \phi_s(\mathbf{x}^{(i)})$ on the target channel only: SMS, $\gamma_1 \sim \mathcal{U}(2, 5)$; silent-call, $\gamma_2 \sim \mathcal{U}(0.3, 0.7)$ with $\text{CALL} \leftarrow \max(0, \gamma_2 \cdot \text{CALL})$; Internet, $\gamma_3 \sim \mathcal{U}(2, 5)$; Blended, all three applied jointly. Non-target channels are left unchanged (silent-call models resource-exhaustion via call suppression [8], [38]).
- 3) Map each constrained triple to a $9 \times 9 \times 3$ grid via $\text{MapToGrid}(\tilde{\mathbf{x}}^{(i)}, B^{(i)})$ with $B^{(i)} \in \mathcal{A}$.

The Phase 1 detectors are then evaluated on the full test split without retraining (results in Sec. VI-B).

D. Phase 3: Adversarial Diffusion Training (ADT)

Phase 3 applies ADT: for each $s \in \mathcal{S} = \{\text{SMS}, \text{Call}, \text{Internet}, \text{Blended}\}$ and round $r = 1, \dots, R$, inverse-guided TabDDPM samples are appended to the training set and the detector is fine-tuned (Algorithm 2). Let $\mathcal{X}_{\text{adv}}^r$ denote the adversarial grids generated in round r and ϕ_{TabDDPM} the Phase 2 TabDDPM denoiser (Supplementary Sec. S-B).

Algorithm 1 Phase 2: TabDDPM Vulnerability Demonstration

Require: Supplementary Algorithms S1–S2 outputs ($\mathcal{D}_{\text{norm}}$, $\mathcal{A}$, attacked-cell set $\mathcal{C}$); detector $\mathcal{M}(\cdot; \theta)$ (written f_θ)

Ensure: Baseline $\{f_{\theta_s^*}\}_s$; $\{F1_{\text{Ph.2}}^{(s)}, \Delta_{F1}^{(s)}\}_s$

- 1: Build $\mathcal{R}_{\text{norm}}$ from normal CDR aggregates; fit StandardScaler on $\mathcal{R}_{\text{norm}}$
- 2: Train TabDDPM p_ψ on scaled $\mathcal{R}_{\text{norm}}$ (Supplementary Sec. S-B); save model and scaler
- 3: **for** each $s \in \{\text{SMS, Call, Internet, Blended}\}$ **do**
- 4: Train f_θ on scenario-specific $\mathcal{D}_s^{\text{train}}$ (Algorithm S2; no TabDDPM in training) $\Rightarrow f_{\theta_s^*}$
- 5: Initialize $\mathcal{X}_a^s$ with multiplier attacks on all $N_{\text{abn}} = 335$ slots (indices 0–166 retained for training)
- 6: **for** $i = 167$ to 334 $\{N_{\text{attack}}^{\text{test}} = 168$ test-attack indices $\}$ **do**
- 7: $\mathbf{x}^{(i)} \leftarrow \text{TabDDPM-Sample}(p_\psi)$ with heuristic scenario steering on target channel(s) during reverse diffusion every $\max(1, \lfloor T/10 \rfloor)$ steps (enumerate item 1; not Phase 3 inverse classifier guidance)
- 8: $\tilde{\mathbf{x}}^{(i)} \leftarrow \phi_s(\mathbf{x}^{(i)})$ {post-sample mild γ ranges in enumerate item 2}
- 9: $B^{(i)} \leftarrow \mathcal{A}[i]$; $\mathcal{X}_a^s[i] \leftarrow \text{MapToGrid}(\tilde{\mathbf{x}}^{(i)}, B^{(i)})$
- 10: **end for**
- 11: Assemble $\mathcal{D}_{\text{test}}^{(s)}$ from 167 held-out normal grids and 168 TabDDPM attack grids $\mathcal{X}_a^s[167:335]$ (335 test images total)
- 12: Evaluate $f_{\theta_s^*}$ on $\mathcal{D}_{\text{test}}^{(s)}$ without retraining; record $F1_{\text{Ph.2}}^{(s)}, \Delta_{F1}^{(s)}$
- 13: **end for**
- 14: **return** $\{f_{\theta_s^*}\}_s, \{F1_{\text{Ph.2}}^{(s)}, \Delta_{F1}^{(s)}\}_s$

1) Inverse Classifier Guidance: During each ADT round, inverse classifier guidance steers TabDDPM sampling *away* from attack-like detector scores. Let $\mathcal{M}(\cdot; \theta_s^r)$ denote the detector *after* ADT round r ($\theta_s^0 = \theta_s^*$ from Phase 1). During generation at round r , inverse guidance uses $\mathcal{M}(\cdot; \theta_s^{r-1})$ (Algorithm 2); we omit subscript s in Eqs. (3)–(5) when s is fixed. For a CDR aggregate x (or its clean estimate $\hat{x}_0$ during reverse diffusion), we form the grid $I_{\text{grid}}(x) = \text{MapToGrid}(x, B)$ by replicating x into all cells of $\mathcal{C} = \{c_1, \dots, c_{40}\}$ on a background grid B (for inverse guidance during ADT sampling, B is drawn from normal training grids; see ADT iterative training below). The scalar attack probability is the mean predicted score over $\mathcal{C}$:

$$p_{\text{attack}} = \frac{1}{|\mathcal{C}|} \sum_{k \in \mathcal{C}} \mathcal{M}(I_{\text{grid}}(x); \theta_s^{r-1})_k. \quad (3)$$

Given target probability $p_{\text{target}} = 0.3$ and guidance scale λ , the guidance term is

$$\nabla_{\text{guidance}} := \lambda \cdot (p_{\text{attack}} - p_{\text{target}}) \cdot \nabla_x p_{\text{attack}}, \quad (4)$$

where $\nabla_x p_{\text{attack}}$ is the gradient of p_{attack} with respect to x ; the guided update is

$$\hat{x}_0 \leftarrow \hat{x}_0 - \nabla_{\text{guidance}}. \quad (5)$$

For non-blended scenarios, gradient components in non-target CDR channels are scaled by 0.7 before applying the update,

Algorithm 2 Phase 3: adversarial diffusion training (ADT)

Require: Phase 1 weights $\{\theta_s^*\}_{s \in \mathcal{S}}$; TabDDPM ϕ_{TabDDPM} (Supplementary Sec. S-B); $(\mathcal{X}_s^{\text{train}}, \mathcal{Y}_s^{\text{train}})$, $\mathcal{X}_s^{\text{test}}, \mathcal{X}_a^{s, \text{Tab}}$ (Phases 1–2); rounds R , guidance λ , samples/round N_{ADT} , fine-tune epochs E_{ADT} , cell set $\mathcal{C}$

Ensure: $\{\theta_s^{R,*}\}_s$; $\{\Xi_s^r\}_{r=0}^R$

- 1: **for** each $s \in \mathcal{S} = \{\text{SMS, Call, Internet, Blended}\}$ **do**
- 2: $\theta_s^0 \leftarrow \theta_s^*$; $\Xi_s^0 \leftarrow \text{Evaluate}(\theta_s^0, \mathcal{X}_a^{s, \text{Tab}})$
- 3: **for** $r = 1$ to R **do**
- 4: $\mathcal{X}_{\text{adv}}^r \leftarrow \emptyset$; $f_{\text{guid}} \leftarrow \max(1, \lfloor T/10 \rfloor)$
- 5: **for** $i = 1$ to N_{ADT} **do**
- 6: TabDDPM reverse diffusion from $x_T \sim \mathcal{N}(0, I)$; at steps $t \bmod f_{\text{guid}} = 0$, $t > 0$, apply inverse guidance (Eqs. (3)–(5)) using $\mathcal{M}(\cdot; \theta_s^{r-1})$
- 7: $x_{\text{adv}} \leftarrow \text{MapToGrid}(\hat{x}_0, B=0)$ {zero background; aggregate in $\mathcal{C}$ only}
- 8: $\mathcal{X}_{\text{adv}}^r \leftarrow \mathcal{X}_{\text{adv}}^r \cup \{x_{\text{adv}}\}$
- 9: **end for**
- 10: Append $\mathcal{X}_{\text{adv}}^r$ to $\mathcal{X}_s^{\text{train}}$ (attack labels) $\Rightarrow (\mathcal{X}_s^{\text{train}, r}, \mathcal{Y}_s^{\text{train}, r})$
- 11: Fine-tune θ_s^{r-1} for E_{ADT} epochs $\Rightarrow \theta_s^r$
- 12: $\Xi_s^r \leftarrow \{\Xi_s^{r, \text{orig}}, \Xi_s^{r, \text{Tab}}\}$ on $\mathcal{X}_s^{\text{test}}, \mathcal{X}_a^{s, \text{Tab}}$
- 13: **end for**
- 14: $\theta_s^{R,*} \leftarrow \theta_s^R$
- 15: **end for**
- 16: **return** $\{\theta_s^{R,*}\}_s, \{\Xi_s^r\}_{r=0}^R$

preventing guidance from distorting unrelated traffic dimensions.

2) ADT Iterative Training Procedure: Round-wise metrics are $\Xi_s^r = \{\Xi_s^{r, \text{orig}}, \Xi_s^{r, \text{Tab}}\}$ on the multiplier test split $\mathcal{X}_s^{\text{test}}$ and TabDDPM test set $\mathcal{X}_a^{s, \text{Tab}}$, respectively; the final weights are $\theta_s^{R,*}$. During inverse guidance, $I_{\text{grid}}(x)$ is formed with a randomly drawn *normal* training background. ADT-appended training grids use a zero background with the aggregate written only into $\mathcal{C}$ (Sec. IV-A). Hyperparameters (R , N_{ADT} , E_{ADT} , λ , p_{target}) are given in Sec. V.

V. EXPERIMENTAL SETUP

Following [4], we use the Telecom Italia Milano CDR collection [35]: a 9×9 subgrid (81 cells), 18 time slots per day (11:00–14:00, 10-minute intervals), yielding 1,116 grid images per scenario split into 781 training (614 normal + 167 attack) and 335 test (167 normal + 168 attack) images. We fix random seed 1 for data shuffling and attack-cell selection ($|\mathcal{C}| = 40$).

TabDDPM is trained on normal aggregate CDR rows (3D features) for 100 epochs (learning rate 10^{-4} , batch size 32). SimpleCNN and ResNet50 are trained for up to 100 epochs in Phases 1–2 (ResNet50: 10% validation split with early stopping to limit overfitting; SimpleCNN: full 100-epoch budget). Phase 3 ADT uses $R = 3$ rounds, $\lambda = 1.0$, $N_{\text{ADT}} = 100$ samples per round and scenario, $E_{\text{ADT}} = 20$ fine-tuning epochs per round, and $p_{\text{target}} = 0.3$. We evaluate Accuracy, Precision, Recall, F1, FPR, and ROC AUC via micro-averaging over $N_{\text{test}} \times 81$ cell decisions ($N_{\text{test}} = 335$). **No normalization** is applied to CNN inputs (consistent with [4]); TabDDPM

TABLE III: Reader’s guide to F1 evaluation protocols.

ID	Protocol	Main location
P1	335-grid TabDDPM test; $\tau = 0.5$; cumulative Phase-3 ADT	Table IV
P2	Same test as P1; fresh 3-round schedule per method; $\tau = 0.5$	Fig. 4 (left)
P3	Locked attack test; $\tau_{F1\max}$ or τ_Y from validation	Fig. 4 (right); Sec. VI-G
P4	Fresh-retrain $\lambda/p_{\text{target}}$ sweep; locked test	Table VII
P5	Multiplier test split; $\tau = 0.5$	Table V

aggregates are standardized with `StandardScaler` fit on normal data only. Implementation details follow Supplementary Algorithms S1–S3 and main-text Algorithms 1–2.

A. Evaluation Protocols for Reported F1

F1 is reported under protocols P1–P5; Table III maps each label to its evaluation setting.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

Extended derivations, sensitivity sweeps, and full result grids are in the Supplementary Material.

A. Evaluation Scope

Our use of “5G” refers to the *deployment context*—CPS services depending on cellular connectivity—rather than a claim the benchmark traces were collected from a 5G standalone (SA) core. We evaluate on the public Milano CDR corpus [35] for reproducible comparison with Hussain *et al.* [4]; coarse CDR aggregates remain generation-agnostic monitoring records used in contemporary 5G anomaly detection [23], [39]. No public labeled per-cell cellular DDoS CDR dataset exists; packet-/flow-level 5G corpora (5G-NIDD [40], NCSRD-DS-5GDDoS [41]) are not directly mappable to our $9 \times 9 \times 3$ grid representation. Empirical claims are scoped to robust CDR-grid detection under realistic synthetic stress tests in a 5G-motivated CPS monitoring setting.

B. Phase Progression Across Diff-DDoS

Table IV summarizes detector performance across phases (all values at $\tau = 0.5$; Ph. 2–3 follow protocol P1 on the TabDDPM test split). Both architectures achieve near-perfect Phase 1 F1 (99.6%–100%) on multiplier-based test attacks—expected because Phase 1 multipliers (Table II) deviate tens to hundreds of σ from normal (Fig. 2). Without retraining, Phase 2 F1 collapses to near zero for SMS, Internet, and Blended on TabDDPM attacks; ResNet50 silent-call F1 falls to 7.6% despite greater model capacity (a comparatively deeper backbone). ADT (Phase 3) recovers ResNet50 F1 to 79.6% (silent-call), 100% (Internet), and 92.8% (Blended) at round 3; SMS F1 stays below 10% at $\tau = 0.5$ (a decision-rule artifact resolved in Sec. VI-G). SimpleCNN silent-call F1 declines under ADT (Sec. VI-H); for these sparse stealth scenarios (SMS flooding and silent-call), ResNet50 is the deployable backbone (Secs. VI-G–VI-H).

TABLE IV: Detector performance across phases (protocol P1; all values in %; deployment-default $\tau = 0.5$). **Ph. 1:** multiplier-based test attacks (Acc, Rec, F1); precision is 100% for all rows (omitted). **Ph. 2:** TabDDPM-generated attacks without retraining. **Ph. 3:** ADT rounds R1–R3. SC: SimpleCNN; R50: ResNet50.

Scenario	Ph. 1			Ph. 2	Ph. 3 F1		
	Acc	Rec	F1		R1	R2	R3
SMS (SC)	99.96	99.85	99.93	0	9.09	9.09	9.09
SMS (R50) ^a	100	100	100	0	6.98	9.74	9.58
Call (SC) ^b	99.85	99.40	99.70	52.63	38.44	12.87	9.71
Call (R50)	99.83	99.32	99.66	7.56	77.22	81.52	79.62
Internet (SC)	100	100	100	0	89.14	96.78	96.62
Internet (R50)	99.80	99.21	99.60	0	100	100	100
Blended (SC)	100	100	100	0	86.55	81.13	79.57
Blended (R50)	100	100	100	0	99.35	97.22	92.79

^aResNet50 SMS values below 10% reflect a fixed- τ artifact (Sec. VI-G).

^bSimpleCNN silent-call Phase 3 decline is interpreted in Sec. VI-H.

C. Attack Realism of TabDDPM-Generated Attacks

We operationalize *realistic* as **manifold-preserving**: synthesized attacks remain within the statistical envelope of normal CDR traffic, modeling stealthy control-plane abuse rather than high-volume volumetric DDoS. We quantify distributional fidelity against the normal CDR aggregate manifold and benchmark against Phase 1 multiplier baselines [4].

For each test attack grid we extract the mean 3D aggregate over the 40 attacked cells (168 grids per scenario) and compare it to a normal reference pool of 5,000 cell-level 3D features from normal-only training grids, using target-channel mean absolute z -score as the primary realism metric and squared maximum mean discrepancy (MMD²) with a radial basis function (RBF) kernel as a distributional check (full definitions in Supplementary Sec. S-C). Separately, 5,000 unconstrained TabDDPM draws yield joint MMD² = 0.50 to that normal pool, confirming generative fidelity before scenario constraints. After mild scenario constraints, TabDDPM attacks deviate by only ≈ 0.6 – 3.5σ on the target channel, whereas multiplier attacks deviate by ≈ 1 – 172σ (Fig. 2; Supplementary Table SI). For silent-call, TabDDPM attacks deviate by only 0.59σ on CALL (target-channel MMD² = 0.19), so Phase 2 traffic is statistically near-normal; this stealth regime helps explain why multiplier-trained ResNet50 F1 collapses to 7.6% on TabDDPM Call grids despite near-perfect Phase 1 multiplier scores. These results confirm TabDDPM as a manifold-preserving stealth-attack synthesizer; limitations are discussed in Sec. VII.

D. ADT Recovery and Multiplier Retention

Beyond Table IV (TabDDPM test, protocol P1), we assess architecture-specific recovery and multiplier-test retention (protocol P5).

SimpleCNN. On the same 335-grid TabDDPM protocol (Table IV, $\tau = 0.5$), Internet (0% $\rightarrow$ 96.6% at R3) and Blended (0% $\rightarrow$ 79.6%) improve markedly. Silent-call F1 drops from 52.6% (Phase 2) to 9.7% at R3—Grad-CAM analysis (Sec. VI-H) shows ADT suppresses attack scores on TabDDPM silent-call samples (mean attack score on attacked cells $0.35 \rightarrow 0.01$) while preserving localization; ResNet50

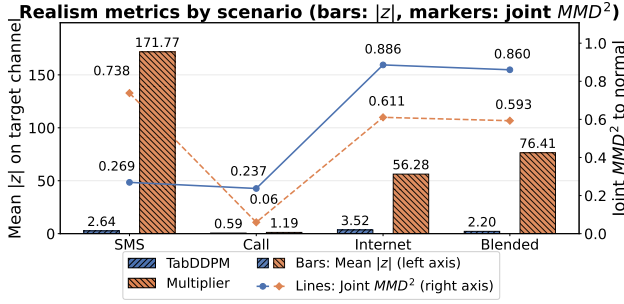


Fig. 2: Dual-axis realism summary on test attack grids (168 per scenario). Bars (left axis) show target-channel mean absolute z -score, and lines (right axis) show joint (3D) MMD^2 to normal, for TabDDPM and multiplier attacks across scenarios. Lower values indicate closer alignment with normal CDR aggregates (higher realism/stealth).

TABLE V: Retention on fixed-multiplier attacks (protocol P5; ResNet50): F1 (%) on the locked multiplier test split at $\tau = 0.5$, before ADT (Phase 1) and after ADT round 3.

Scenario	Phase 1 (pre-ADT)	ADT round 3
SMS	100.00	100.00
Call	99.66	100.00
Internet	99.60	66.80
Blended	100.00	100.00

Call F1 still rises to 79.6%. SMS F1 remains near 9% throughout.

Cell-level localization. Per-cell accuracy heatmaps for the Blended scenario before and after ADT (Fig. 3) show that ADT converts a patchwork of inconsistent per-cell accuracies into a near-uniform 9×9 accuracy grid ($\approx 100\%$ per cell), resolving spatial inconsistencies across attacked locations.

Retention on multiplier-based tests. Table V shows that ResNet50 retains 100% F1 on SMS, silent-call, and Blended on the *multiplier* test split at $\tau = 0.5$ after ADT. For Internet, F1 decreases from 99.6% to 66.8%, while attack-cell recall remains 100% (TP= 6720, FN= 0), indicating a calibration shift driven by false positives on normal cells rather than loss of multiplier-attack recognition; increasing τ to 0.99 raises Internet F1 to 86.3%. SimpleCNN retains 100% F1 on all four scenarios after ADT round 3.

E. Comparison with Generative and Adversarial-Training Baselines

We compare ADT against a fixed-multiplier-trained detector without iterative retraining [4], fast gradient sign method (FGSM) adversarial training, projected gradient descent (PGD) adversarial training (ℓ_∞ , $\varepsilon = 0.1 \times$, 10 steps), and conditional tabular GAN (CTGAN) [27] (150 epochs on up to 20,000 normal cell-level CDR features, same mild scenario scaling as Phase 2). All iterative methods follow the Phase 3 schedule (3 rounds, 100 augmented grids/round, 20 fine-tuning epochs/round) on ResNet50, evaluated on the locked TabDDPM test set (the same 335-image split as protocol P1: 167 normal + 168 TabDDPM attack grids; indices fixed after Phase 2 generation). We report $\tau = 0.5$ as the deployment-default (attack-type labels are unavailable at deployment time) before presenting validation-calibrated results with per-method

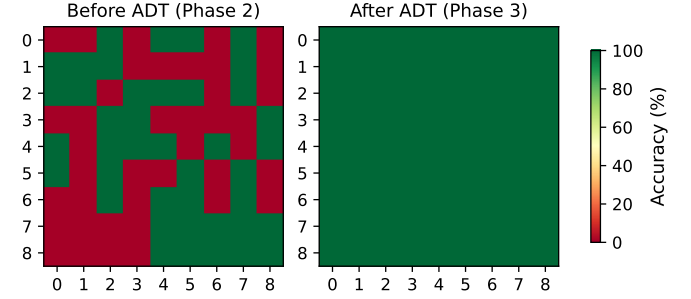


Fig. 3: Cell-level accuracy heatmaps for the Blended scenario before and after ADT.

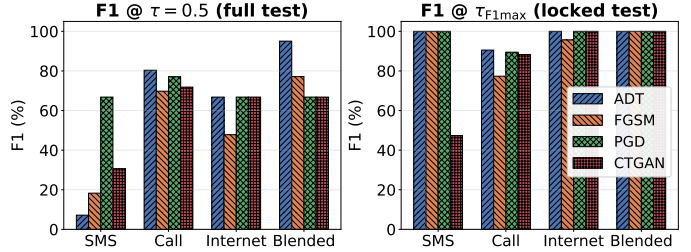


Fig. 4: Baseline comparison on the locked TabDDPM test set (ResNet50, round 3). **Left (P2):** F1 at deployment-default $\tau = 0.5$ (FGSM, PGD, and CTGAN each follow the identical 3-round schedule used for ADT). **Right (P3):** F1 at τ_{F1max} (validation-selected; Sec. VI-G). Compare methods within each panel; each panel applies a single threshold rule. Numeric values appear in Supplementary Table SIII.

τ_{F1max} (Sec. VI-G). Fig. 4 summarizes round-3 F1 under protocols P2 (left) and P3 (right); compare methods within each panel, as the two panels apply distinct threshold rules (common $\tau = 0.5$ versus per-method τ_{F1max}).

Results at $\tau = 0.5$ (iterative protocol). Fig. 4 (left panel) and Supplementary Table SIII report round-3 F1 under protocol P2 (iterative baseline-comparison: FGSM, PGD, and CTGAN each follow the same 3-round retraining schedule as ADT). Under this protocol ($\tau = 0.5$, locked TabDDPM test), ADT is the only method reaching operational F1 on **Blended** without per-method tuning (95.1% vs. $\leq 77.2\%$ for FGSM/PGD/CTGAN). On **Call**, ADT leads (80.4%; next-best PGD 77.0%). On **Internet**, ADT ties PGD/CTGAN (66.8%). On **SMS**, PGD reports higher round-3 F1 at $\tau = 0.5$ (P2: 66.8% vs. ADT 7.2%); on the locked attack test split both achieve AUC= 1.0—a decision-rule artifact resolved in Sec. VI-G (Fig. 4, right panel).

Validation-calibrated comparison. Fig. 4 (right panel) reports τ_{F1max} selected on the held-out attack validation split and applied once to the locked attack test (P3; distinct from the 335-grid $\tau = 0.5$ scores in Table IV). ADT and PGD are essentially tied on **Call** (90.5% vs. 89.5%; bootstrap 95% CIs [85.6, 94.7] and [83.3, 94.3], overlapping). The clearest gap is **SMS**: ADT/FGSM/PGD reach 100% while CTGAN remains at 47.3%. Internet and Blended approach 100% for all methods after calibration.

Is diffusion strictly necessary? FGSM and PGD are training-time perturbation methods that distort existing grids but cannot synthesize standalone manifold-consistent attack samples for independent stress-testing; PGD is competitive on

TABLE VI: Offline generative and augmentation cost (RTX 4090).

Method	Train (s)	Aug. 100
FGSM / PGD	—	$\lesssim 0.01$
CTGAN	62.5	0.007
TabDDPM	1296.0	—
ADT round	—	21.7

Note: FGSM/PGD: no generative model; augmentation is on-the-fly grid perturbation. CTGAN and TabDDPM: one-time offline training. ADT round: inverse-guided TabDDPM generation plus 20-epoch detector fine-tuning per round.

TABLE VII: Inverse-guidance sensitivity summary (ResNet50, round-3 F1 %). The operational default row reports P1 (cumulative Phase-3 ADT on the full 335-grid TabDDPM test, Table IV); sweep rows report a fresh-retrain, locked-test one-at-a-time sweep. The full grid is in Supplementary Table SII.

Protocol	λ	p_t	Blended	SMS	Internet
Operational default	1.0	0.3	92.8	9.6	100.0
	1.0	0.3	66.8	100.0	66.8
Fresh-retrain sweep	2.0	0.3	66.8	100.0	100.0
	1.0	0.1	100.0	100.0	65.9

Call but lacks CDR aggregate realism (Sec. VI-C). FGSM and PGD require no generative pre-training, whereas CTGAN trains $\approx 21\times$ faster than TabDDPM (62.5s vs. 1296.0s; Table VI), but showed a large Blended round-wise regression (99.7% $\rightarrow$ 66.8% at $\tau = 0.5$) and calibrated SMS F1 of only 47.3% (P3; Supplementary Table SIII bottom block). ADT combines manifold-preserving synthesis with deployable Blended robustness at the default threshold—properties not jointly matched by the alternatives.

F. Inverse-Guidance Sensitivity and Online Latency

ADT depends on guidance scale λ (steering strength) and p_{target} (adversarial difficulty). We perform a one-at-a-time sweep on ResNet50: $\lambda \in \{0.25, 0.5, 1.0, 2.0\}$ at $p_{\text{target}}=0.3$, and $p_{\text{target}} \in \{0.1, 0.3, 0.5, 0.7\}$ at $\lambda=1.0$. Target-channel MMD² stays flat (≈ 0.51 – 0.57), so neither hyperparameter materially affects on-manifold realism. On the fresh-retrain one-at-a-time sweep (P4; locked TabDDPM attack test, round-3 F1 at $\tau = 0.5$) in Table VII, robustness is *sensitive and non-monotonic*: Internet F1 rises from 66.8% at $\lambda=1.0$ to 100% at $\lambda=2.0$; Blended F1 rises from 66.8% at $p_{\text{target}}=0.3$ to 100% at $p_{\text{target}}=0.1$. The table also reports the operational Phase-3 default; the full sweep is in Supplementary Sec. S-D. (1.0, 0.3) is the fixed deployment compromise, not a global optimum.

Runtime CPS monitoring requires only the online forward-pass latencies in Table VIII. Diff-DDoS targets *periodic* CDR grid analytics, not per-packet or Ultra-Reliable and Low-Latency Communications (uRLLC)-level (~ 1 ms) inference: each $9 \times 9 \times 3$ grid aggregates a **10-minute** window (Sec. V), so the operative CPS latency budget is the inter-grid interval (600 s). SimpleCNN attains 35.7 ms mean (49.5 ms p95) and ResNet50 42.7 ms mean (60.4 ms p95) on an RTX 4090 reference GPU; worst batch-1 p95 is $\approx 10^4\times$ below the aggregation window (Supplementary Table SVIII). Offline pipeline cost is in Supplementary Sec. S-H (Table SVI).

TABLE VIII: Online single-grid inference latency for Phase 3 (ADT) detectors (batch = 1, scenario-averaged; RTX 4090, 500 iterations after 50 warmup). The per-scenario and batch = 32 breakdown is in Supplementary Table SVIII.

Arch.	Mean (ms)	p95 (ms)	Grids/s
SimpleCNN	35.7	49.5	28.1
ResNet50	42.7	60.4	23.4

ROC Curves (Micro-averaged, Blended scenario)

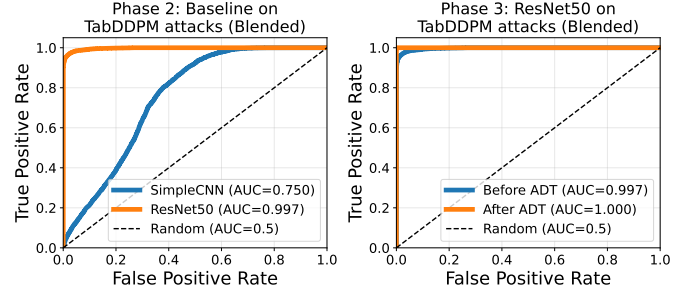


Fig. 5: ROC curves for the Blended scenario on TabDDPM-generated attacks. Left: Phase 2 comparison of SimpleCNN and ResNet50. Right: ResNet50 before and after ADT in Phase 3.

G. Operating Points: ROC-AUC and Threshold Calibration

Micro-averaged ROC AUC (full table in Supplementary Table SVII; representative Blended ROC curves in Fig. 5, with additional ROC/PR curves in the Supplement) confirms Phase 1 perfect separability (AUC= 1.0). Phase 2 reveals vulnerability: SimpleCNN drops to AUC= 0.43 (SMS) and 0.75 (Blended), while ResNet50 holds 0.90–0.99. After ADT, SimpleCNN recovers to 0.85 (SMS) and 0.96 (Blended); ResNet50 reaches 0.95–1.0. The AUC/F1 gap is a decision-rule artifact: for SMS the detector correctly ranks attacks (ResNet50 AUC up to 1.0 after ADT) yet $\tau = 0.5$ yields near-zero SMS F1 (Table IV; locked attack split in Fig. 6) because attack scores concentrate below 0.5; validation-based threshold calibration (below) resolves this.

We implement validation-based threshold calibration and cost-sensitive retraining for the two threshold-sensitive sparse-attack scenarios (SMS flooding and silent-call). For each, τ_{F1max} (validation F1-maximizing) and τ_Y (Youden's $J = \text{TPR} - \text{FPR}$) [42] are selected on a held-out validation split of the TabDDPM *attack-only* test grids (84 of 168 attack images, 50% split, seed 1) and applied once to the remaining 84 locked attack test grids (separate from the primary 335-image micro-averaged evaluation in Table IV), so no test-set tuning occurs. Fig. 6 compares locked-test F1 at $\tau = 0.5$, τ_{F1max} , and τ_Y for Phase 2 (TabDDPM, no ADT) versus Phase 3 (ADT) on SMS flooding and silent-call. For ResNet50 on SMS flooding, calibration alone resolves the apparent ADT failure: on the locked attack split (Fig. 6), F1 rises from 4.82% at $\tau = 0.5$ to 99.55% at $\tau_{\text{F1max}} = 0.05$ (AUC= 1.0, FPR= 0%; Table SIV), confirming that ADT learned the attack and that the sub-10% SMS F1 at $\tau = 0.5$ in Table IV was a fixed-threshold artifact rather than a learning failure.

We also evaluate class-weighted BCE with positive-class

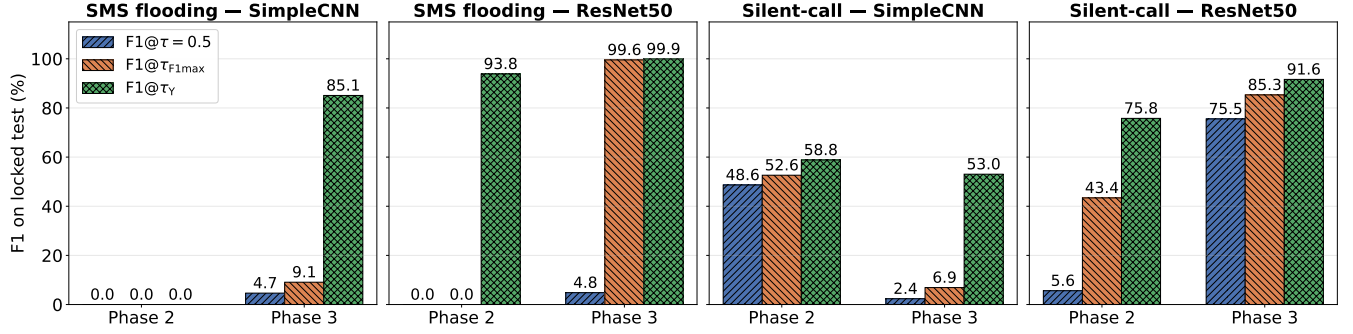


Fig. 6: Locked-test F1 at the default threshold versus validation-calibrated thresholds (τ_{F1max} , τ_Y) for the SMS and silent-call scenarios.

weight ($w_+ \approx 8.5$) and focal loss ($\gamma = 2$, $\alpha = 0.75$) [43] over five seeds. For SMS flooding, neither calibration nor cost-sensitive retraining raises SimpleCNN F1 above $12.3 \pm 8.6\%$ (large seed variance), indicating a capacity limit on the lightweight backbone; we therefore adopt ResNet50 as the deployable detector for sparse attacks. For silent-call, weighted BCE raises mean locked-test AUC from 0.65 to 0.85 ± 0.09 , but F1 at τ_{F1max} remains only $48.5 \pm 25.3\%$ with large seed variance. The full per-architecture grid, Youden thresholds with FPR, and cost-sensitive comparisons appear in Supplementary Sec. S-F (Table SIV); see also Fig. 6.

H. Grad-CAM Interpretability of the SimpleCNN Silent-Call Anomaly

SimpleCNN silent-call F1 falls from 52.6% (Phase 2) to 9.7% (Phase 3, Table IV), which appears to contradict the ADT robustness claim. We apply Gradient-weighted Class Activation Mapping (Grad-CAM) [44] to SimpleCNN’s last convolutional layer on three grid types—normal (real) CDR snapshots, TabDDPM silent-call attacks, and fixed-multiplier baseline attacks—comparing the Phase 2 detector with the Phase 3 ADT model over $N=40$ test grids per condition. Warmer overlay colors in Fig. 7 indicate stronger Grad-CAM activation (where the model was most sensitive on that grid image); $\hat{p}$ is the mean predicted attack score on attacked cells, analogous to p_{attack} in Sec. IV-D. The figure shows three illustrative TabDDPM silent-call grids in which Phase 2 assigns moderate $\hat{p}$ on attacked cells while Phase 3 drives $\hat{p}$ near zero, even though the fraction of Grad-CAM mass on attacked cells (attn) stays $\approx 50\%$ in both phases.

Fig. 8 summarizes the score–saliency asymmetry quantitatively: across $N=40$ TabDDPM silent-call test grids, mean attacked-cell scores drop from 0.35 ± 0.47 (Phase 2) to 0.01 ± 0.07 (Phase 3), matching the F1 collapse, while saliency mass on attacked cells stays comparable (0.51 ± 0.01 vs. 0.51 ± 0.02). On multiplier-based attacks, mean scores remain near 1.0 in both phases, while Phase 3 saliency becomes diffuse (0.18 ± 0.24 vs. 0.51 ± 0.01 ; Supplementary Table SV). ADT on SimpleCNN therefore overfits to stealth guidance samples and *suppresses* detection on manifold-preserving TabDDPM traffic rather than losing localization; ResNet50 still reaches 79.6% F1 under the same protocol. Full numerical Grad-CAM results are in Supplementary Sec. S-G (Table SV).

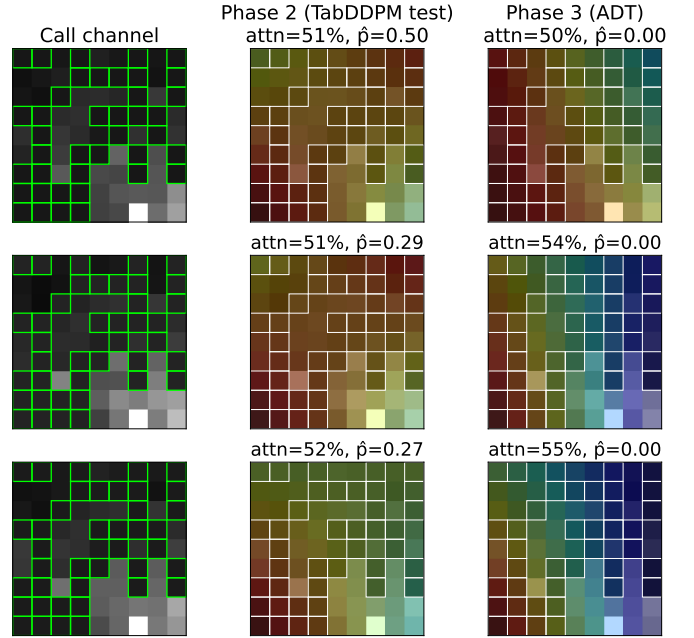


Fig. 7: Grad-CAM overlays for TabDDPM silent-call attacks (SimpleCNN): Call-channel input (left) and saliency maps for Phase 2 (trained on multiplier attacks, tested on TabDDPM silent-call grids) vs. Phase 3 ADT (same backbone after retraining). Boxed cells mark ground-truth attacked locations (green on the input panel, white on the overlays). Subplot titles report the mean predicted attack score on attacked cells ($\hat{p}$) and the fraction of Grad-CAM saliency mass on those cells (attn).

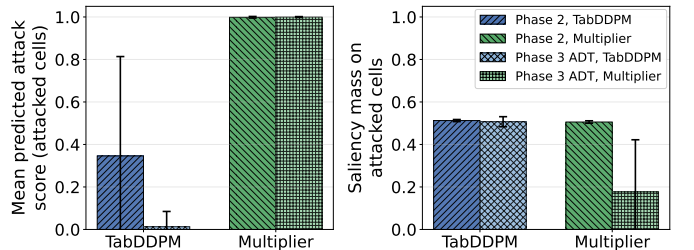


Fig. 8: Quantitative Grad-CAM summary for SimpleCNN before vs. after ADT on attacked cells. *Left*: mean predicted attack score on attacked cells for TabDDPM and multiplier attack sets. *Right*: Grad-CAM saliency mass on attacked cells for the same sets. Bars compare Phase 2 and Phase 3 (ADT); error bars denote ± 1 sample standard deviation over $N = 40$ test grids per condition.

VII. LIMITATIONS AND CONCLUSION

We found that multiplier-trained CDR detectors fail on manifold-preserving attacks: Phase 2 F1 collapsed to near zero in three of four scenarios (Table IV), while ADT with ResNet50 recovered deployable F1 for silent-call (79.6%), Internet (100%), and Blended (92.8%), retaining near-perfect multiplier-test performance except an Internet calibration shift (Table V). This extends Hussain *et al.* [4] from multiplier baselines to manifold-preserving stress tests and applies inverse-guided TabDDPM for iterative cellular IDS hardening. Calibration resolves the SMS threshold artifact (Sec. VI-G), Grad-CAM shows SimpleCNN can suppress stealth silent-call scores without losing localization (Sec. VI-H), and ADT matched or exceeded gradient- and GAN-based baselines (Sec. VI-E).

Three limitations bound external validity. No labeled per-cell cellular DDoS CDR ground truth exists (Sec. VI-A), so evaluation uses synthetic rather than operator-confirmed attack traces. Attack realism uses coarse 3D aggregate statistics—mean $|z|$ and MMD² on SMS/CALL/INTERNET sums (Sec. VI-C)—a proxy that omits full grid spatiotemporal structure and does not establish field adversary fidelity. Finally, the Milano corpus reflects 3G/4G-era metropolitan traffic [35], and ten-minute grid monitoring targets periodic CPS analytics rather than sub-millisecond uRLLC control (Sec. VI-F).

Operator CDR validation and packet-level 5G corpora where mapping permits [40], [41] should confirm ADT transfer beyond Milano; future work includes adaptive adversaries and regional scale-up [6]. For 5G-connected CPS and IoT services, stealthy signaling abuse threatens network and physical-process availability; pre-deployment stress testing is a practical safeguard. Manifold-preserving TabDDPM attacks expose multiplier-trained brittleness, and ADT recovers robustness with ResNet50 and calibration, offering a workflow to harden cell-level DDoS monitors when labeled attack CDRs remain scarce.

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